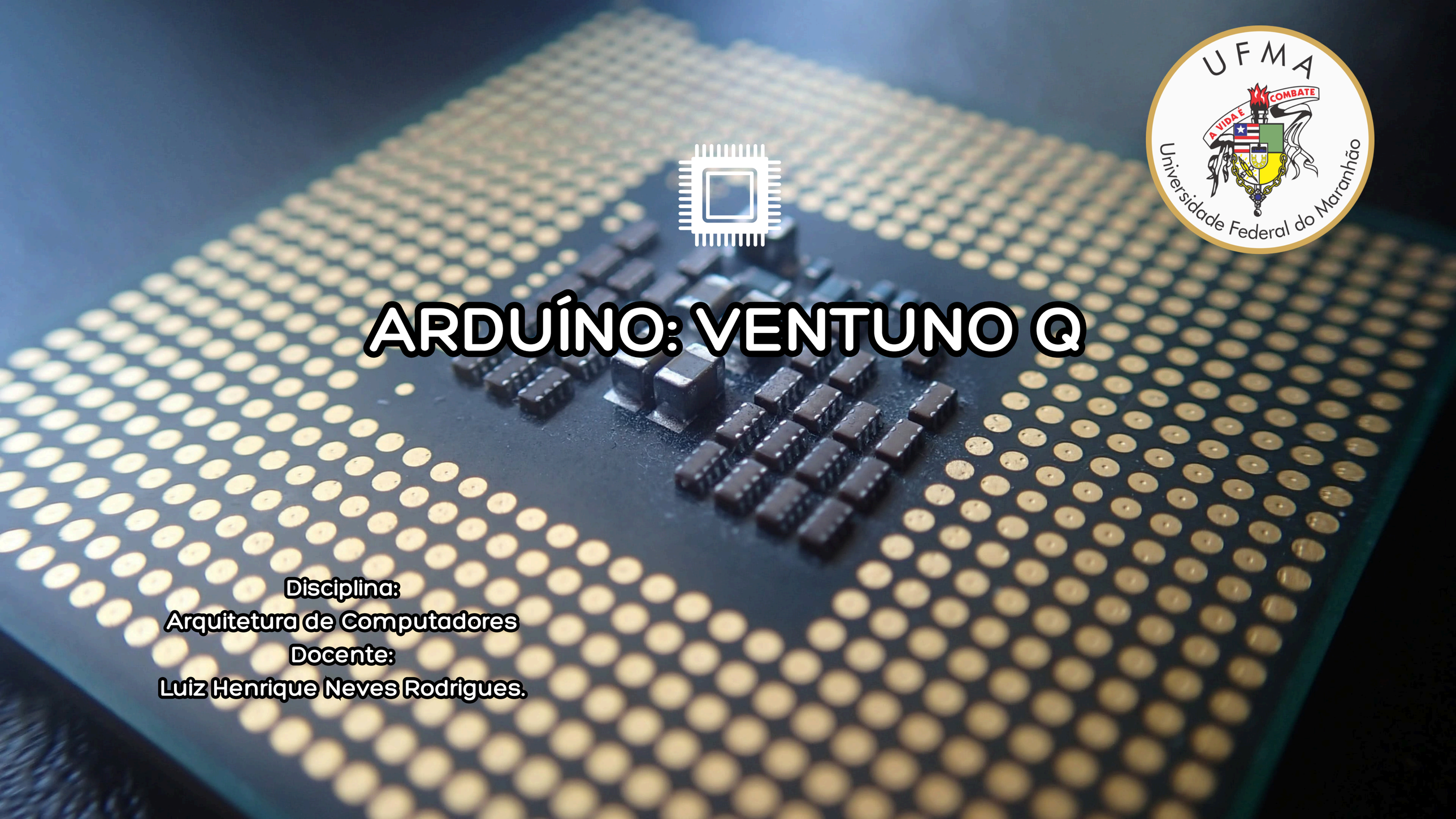


ARDUÍNO: VENTUNO Q

Disciplina:
Arquitetura de Computadores
Docente:
Luiz Henrique Neves Rodrigues.

INTEGRANTES

- Krislayne Rawane Silva Vieira;
- Lucas Santos Baldez;
- Pedro Santos Baldez.

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1 - Introdução

2 - Microcontrolador

3 - Microprocessador

4 - Conclusão

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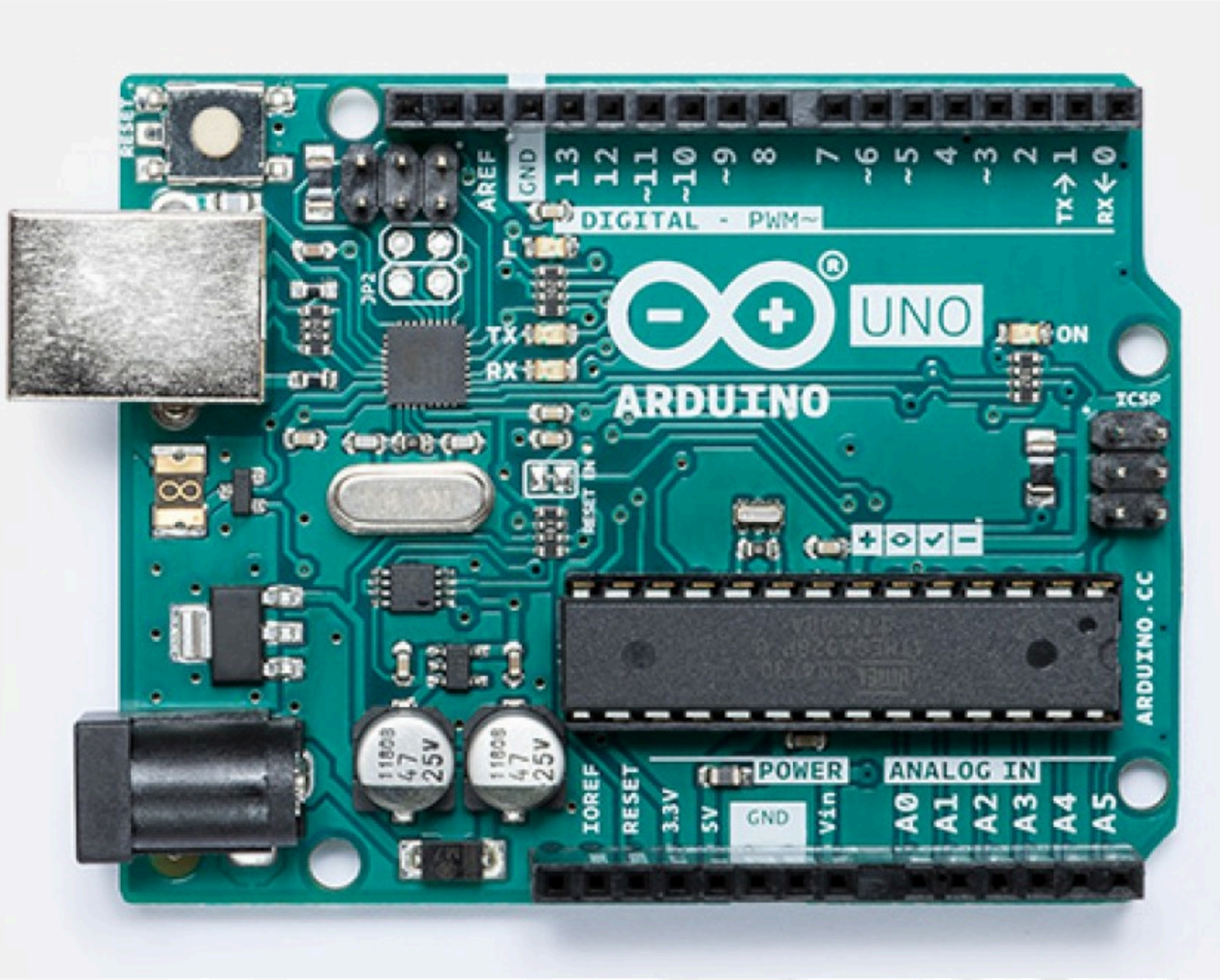
1 - Introdução

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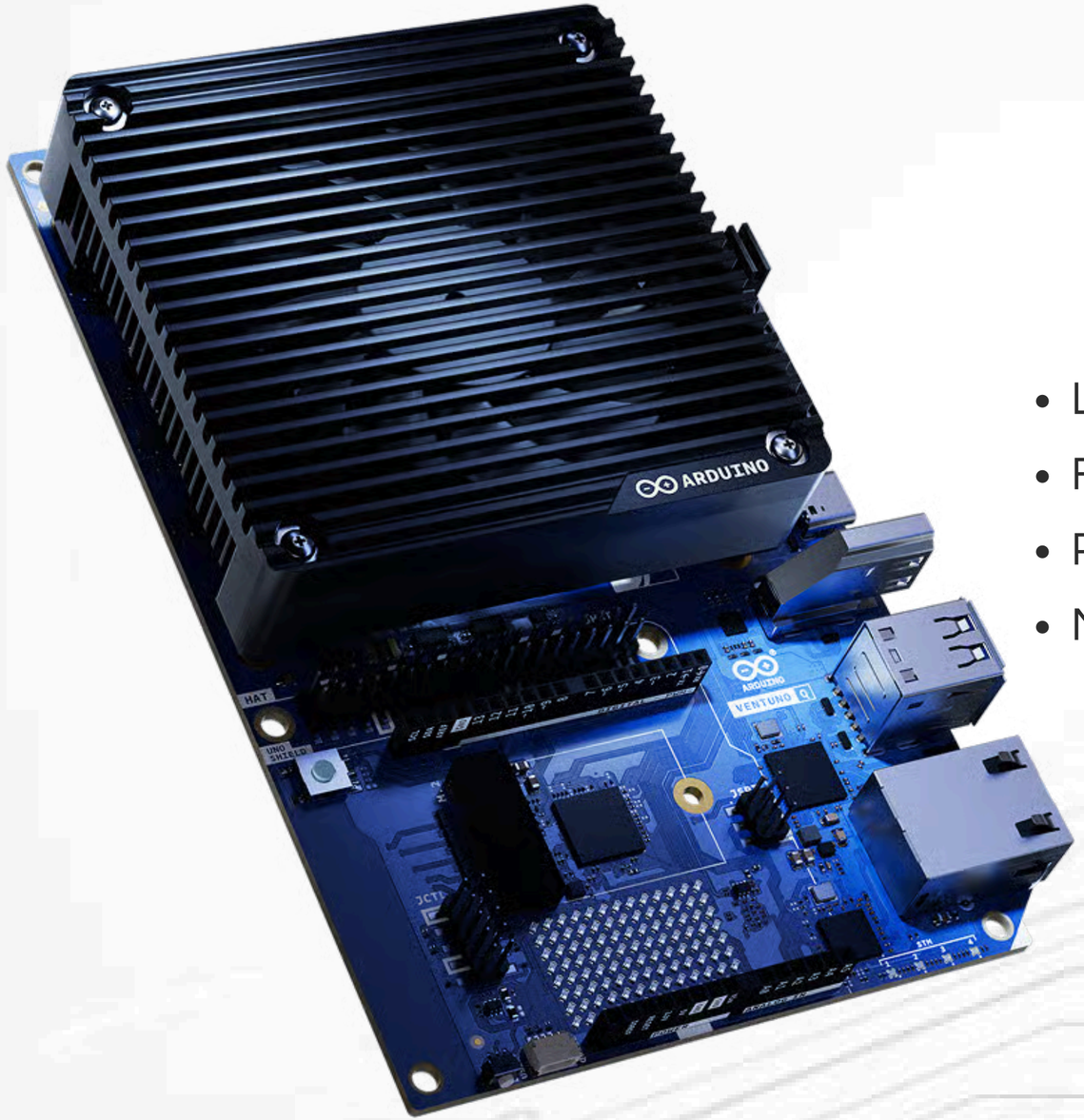
4 - Conclusão

ARDUÍNO



- É UM MINI COMPUTADOR EMBARCADO;
- USADO EM PROJETOS DE ELETRÔNICA E ROBÓTICA;
- PROGRAMÁVEL UTILIZANDO LINGUAGEM C/C+

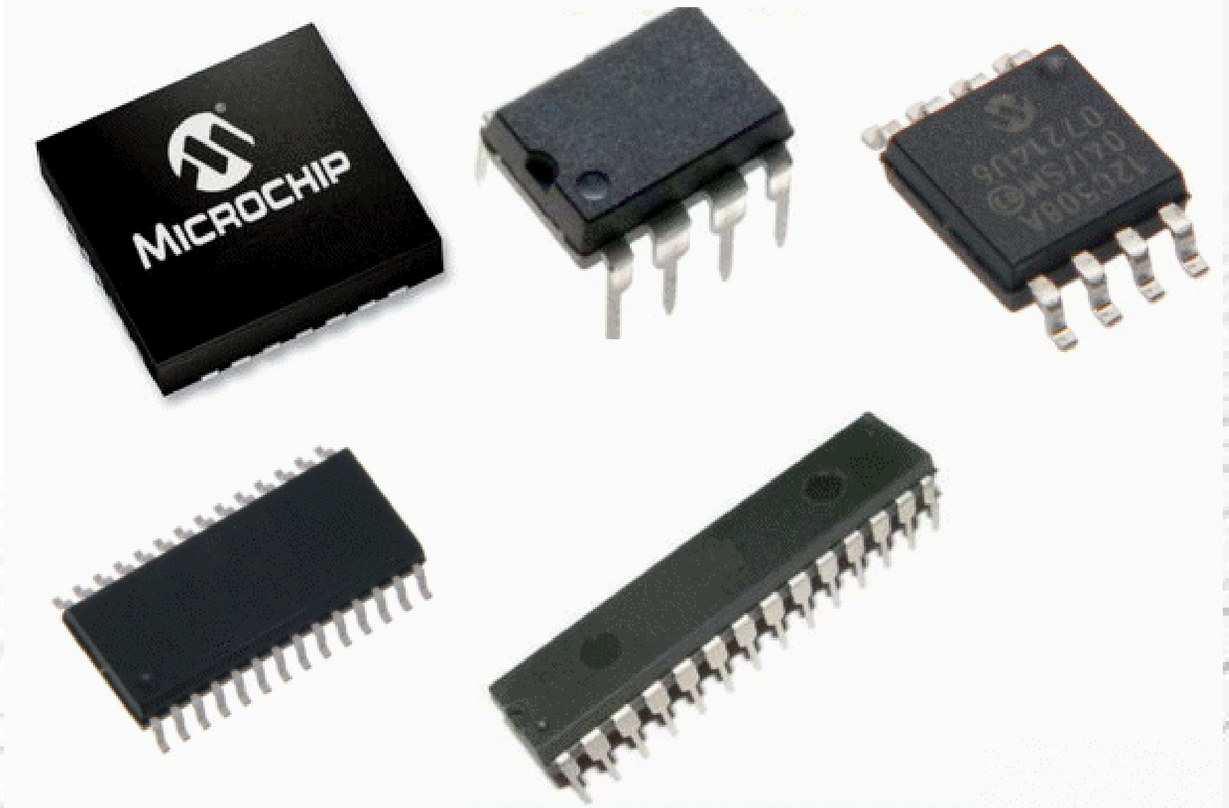
ARDUÍNO: VENTUNO Q



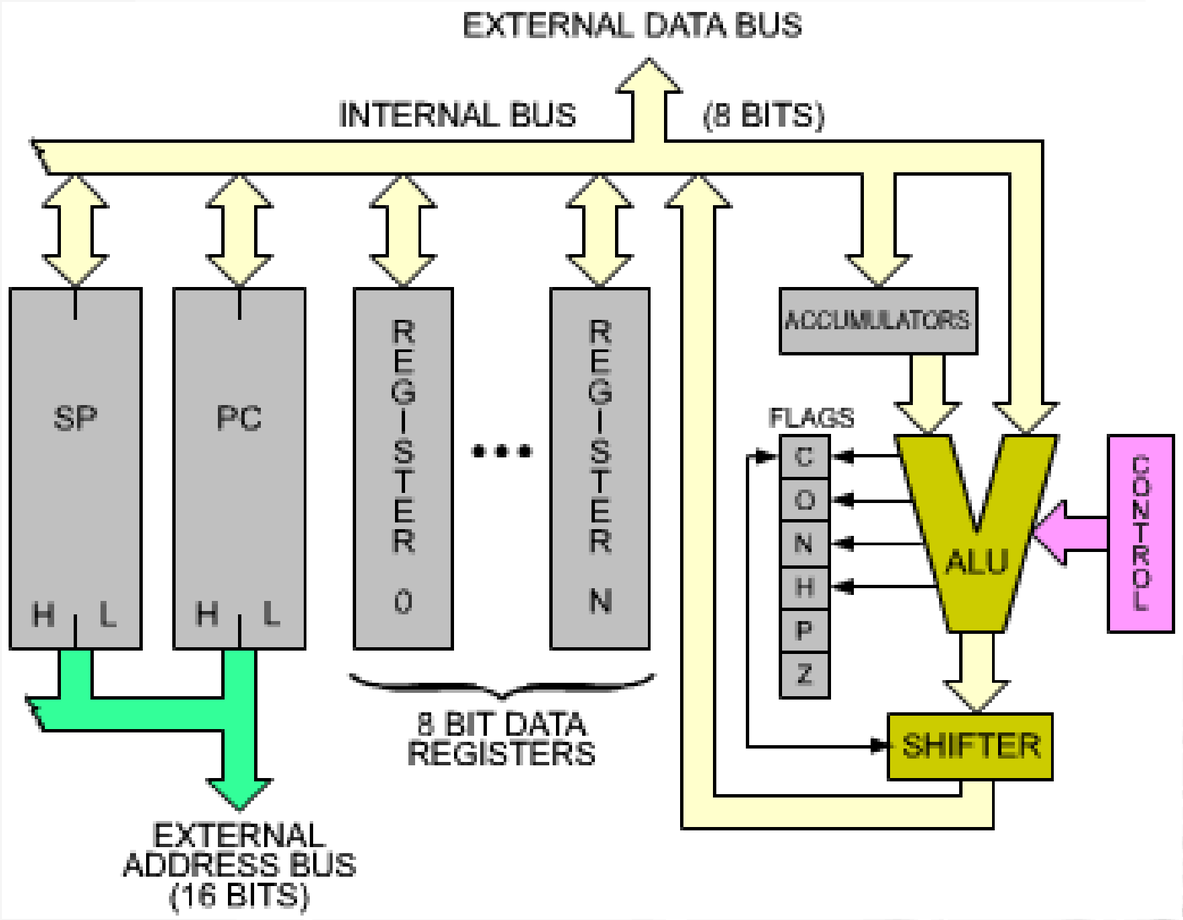
- LANÇADO EM 25 DE AGOSTO DE 2026;
- FOCO EM APLICAÇÕES QUE UNEM IA COM ROBÓTICA;
- PROCESSADOR: DRAGONWING IQ8;
- MICROCONTROLADOR: STM32H5.

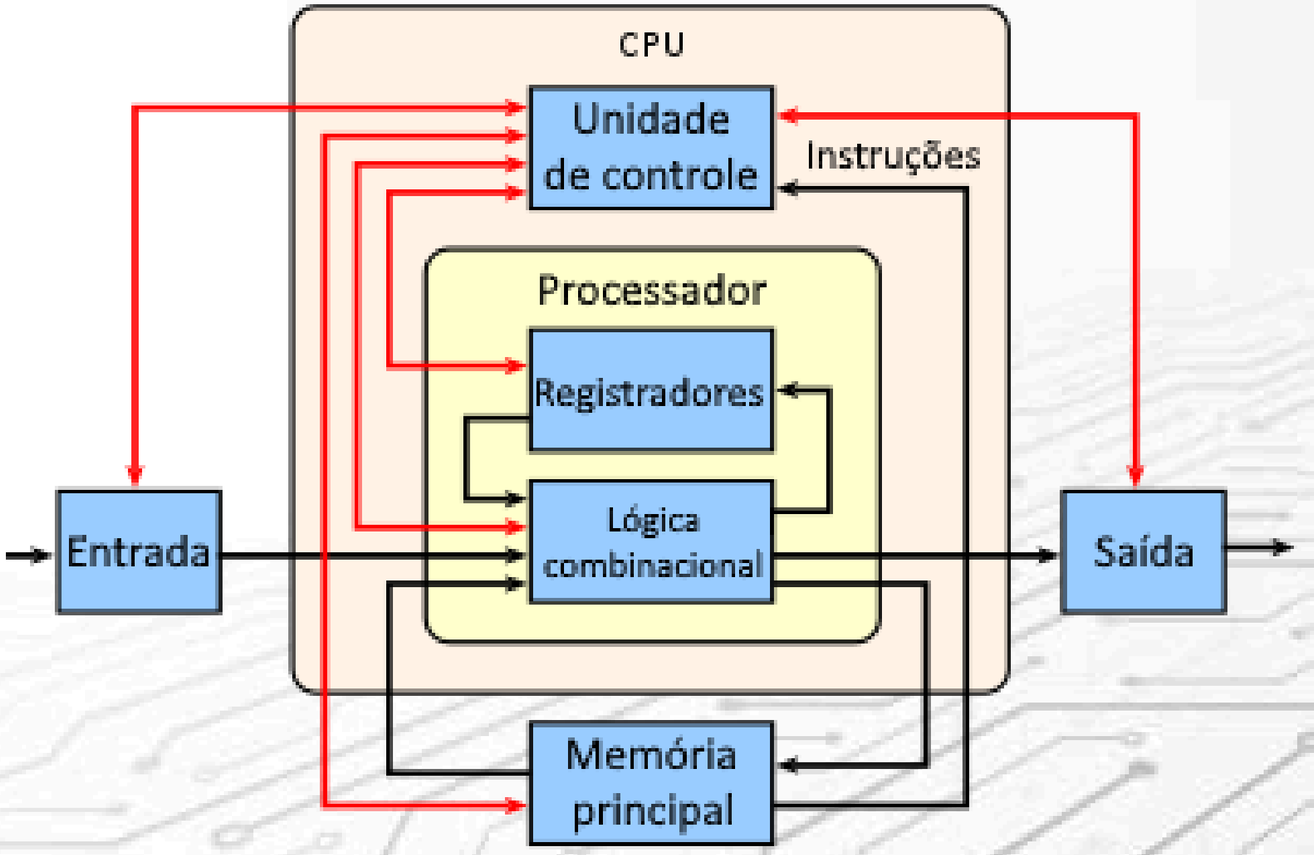
MICROPROCESSADOR
X
MICROCONTROLADOR



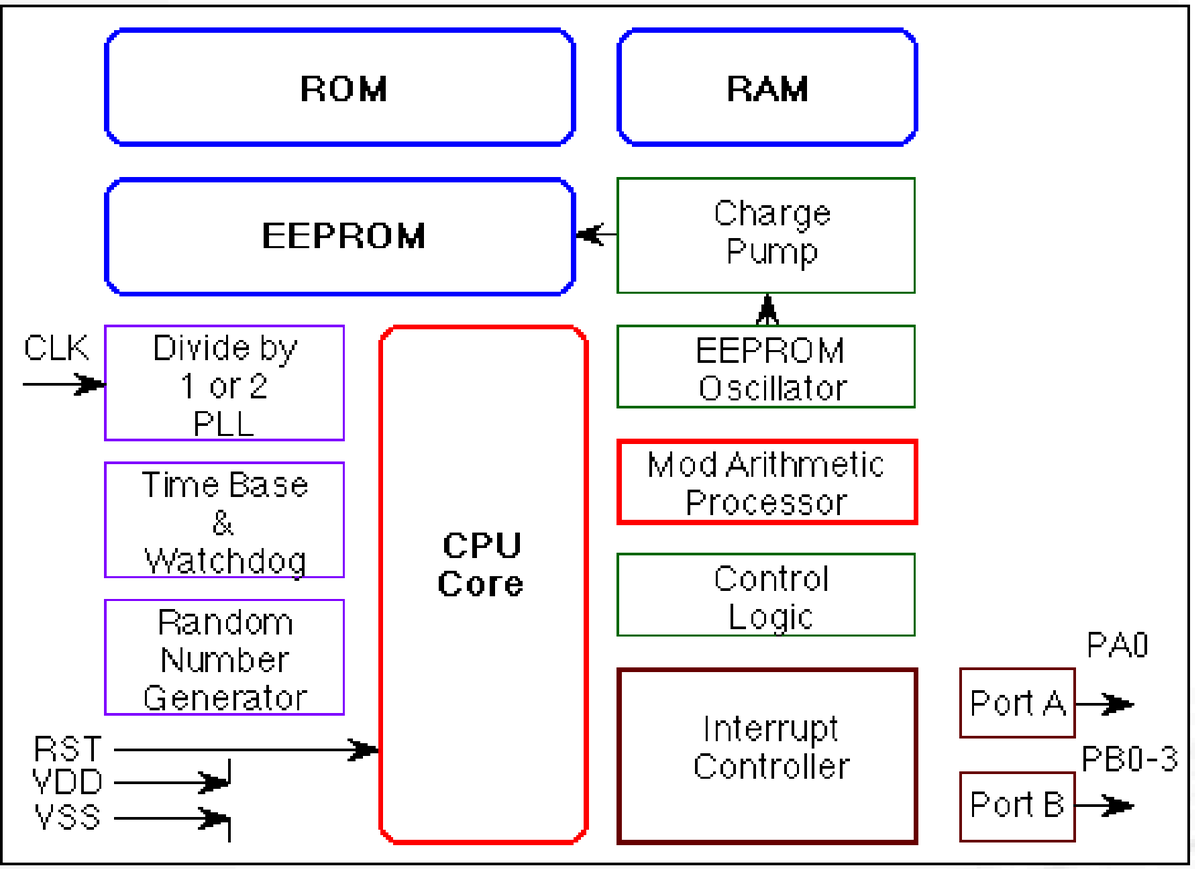


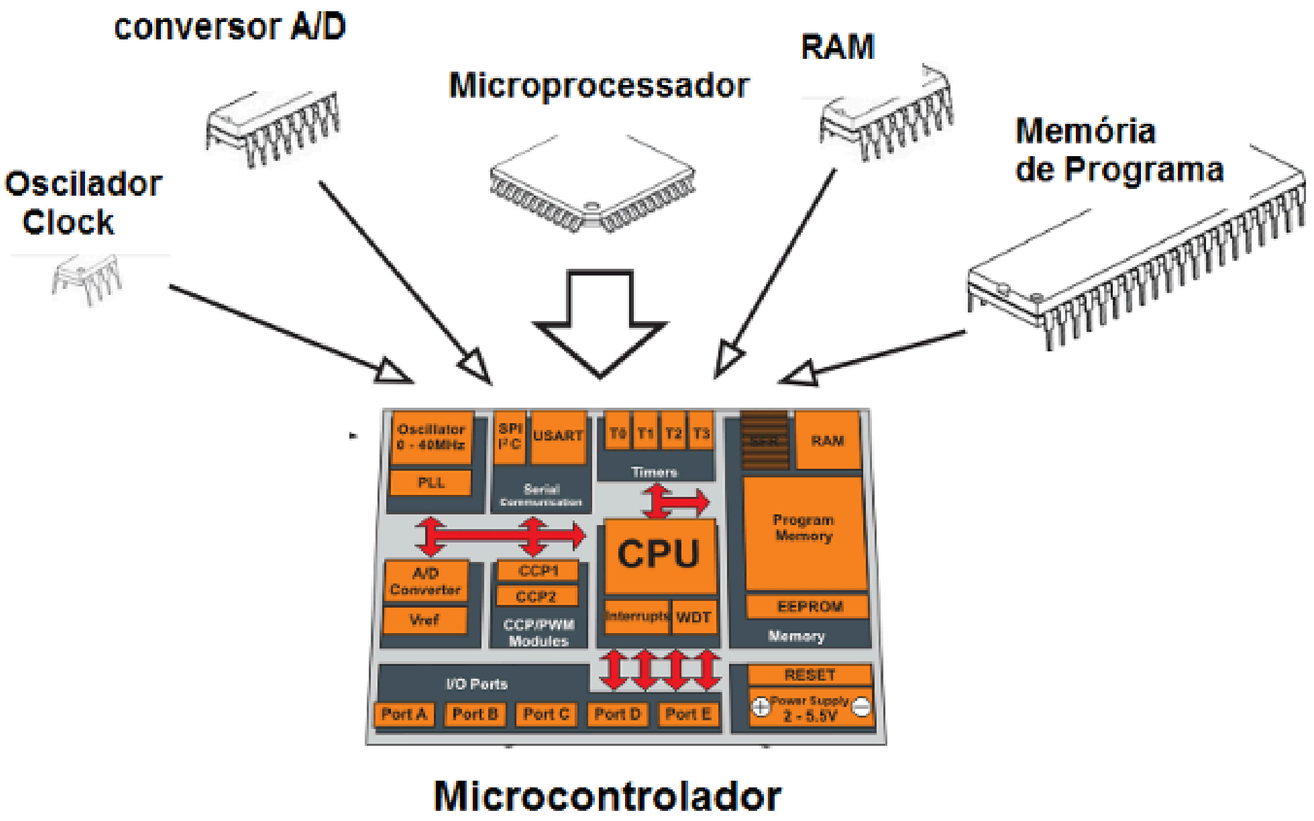
MICROPROCESSADOR





MICROCONTROLADOR





Microcontrolador

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MICROCONTROLADOR

STM32H5F5:

- **ARM® CORTEX® M33 AT 250MHZ**
- **4MB FLASH**
- **1.5MB RAM**

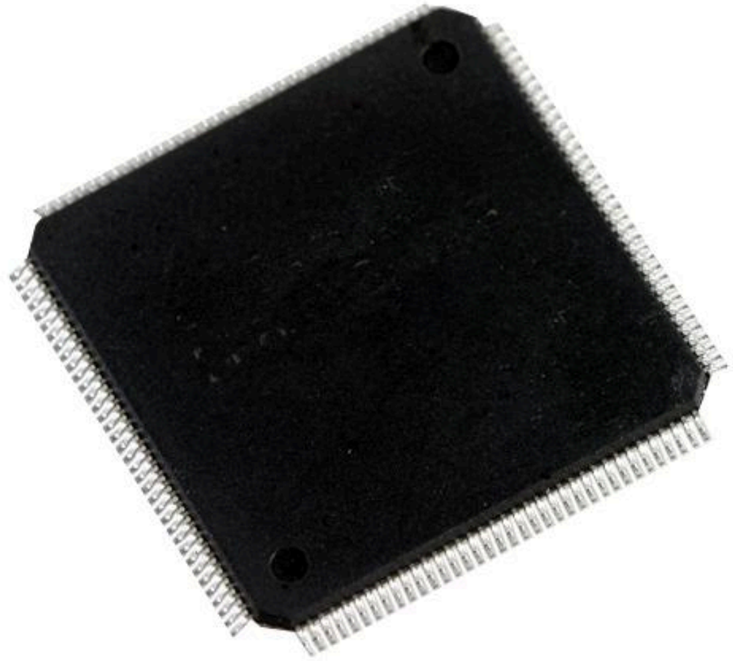
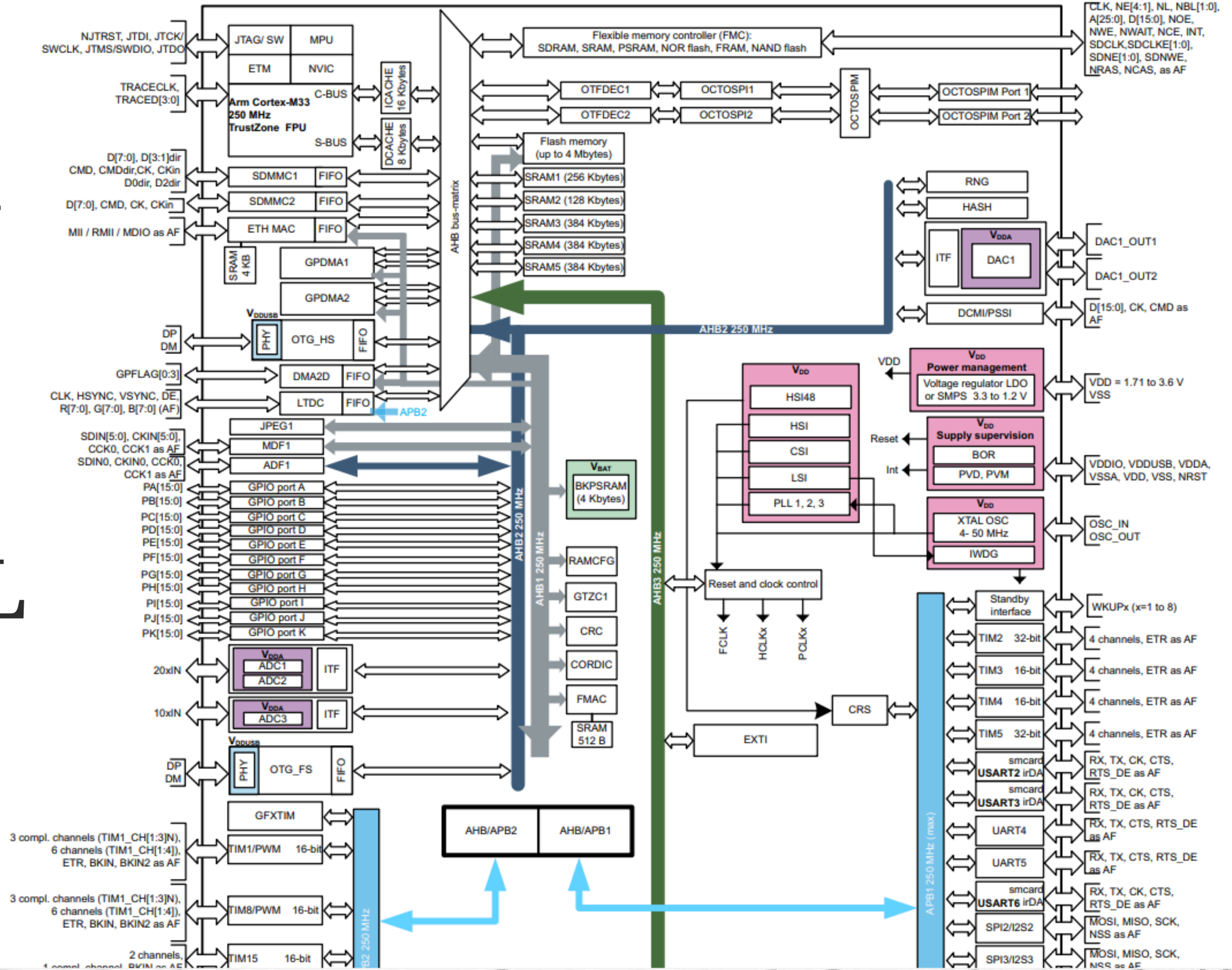


DIAGRAMA DE BLOCO DE ALTO NÍVEL



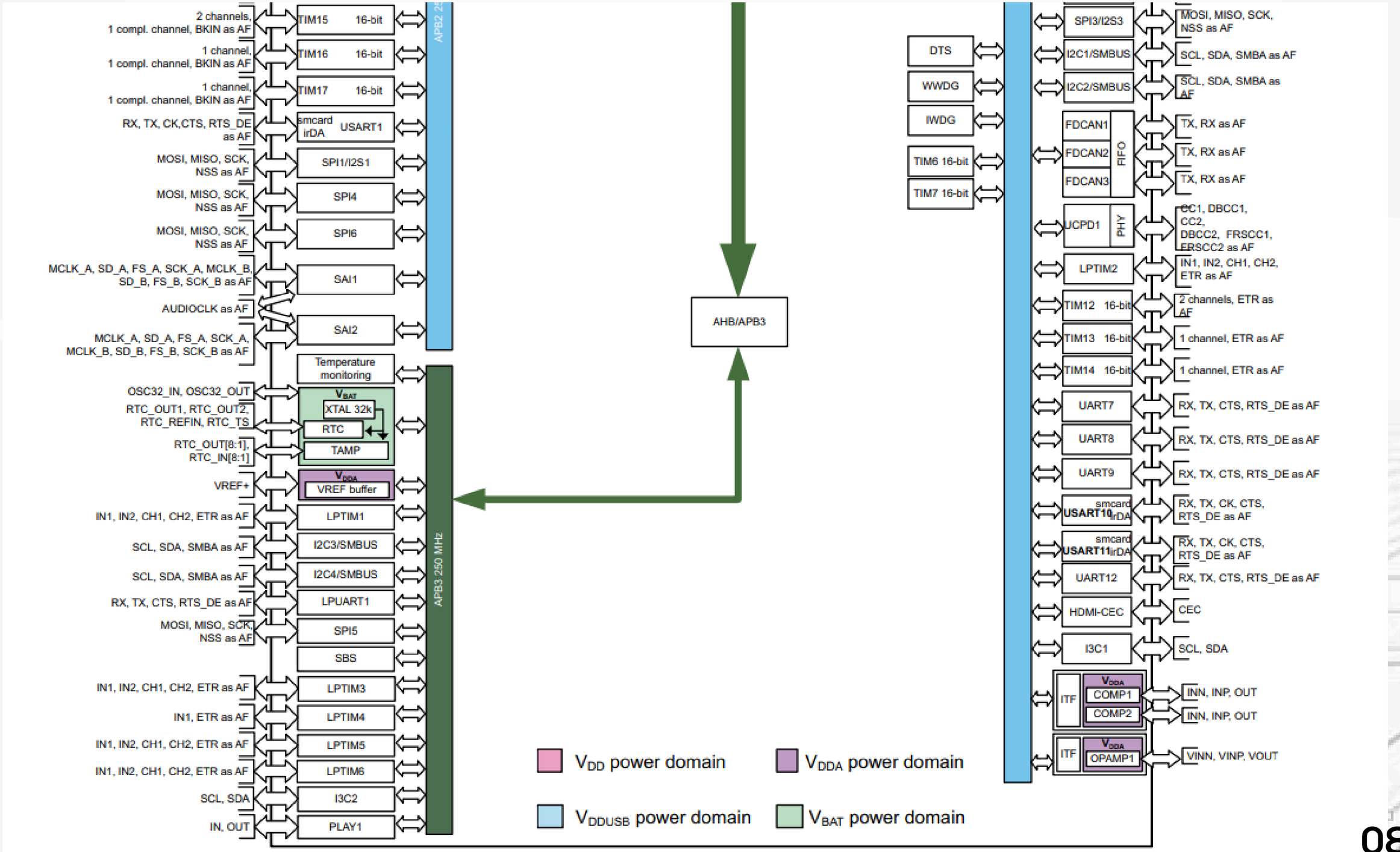
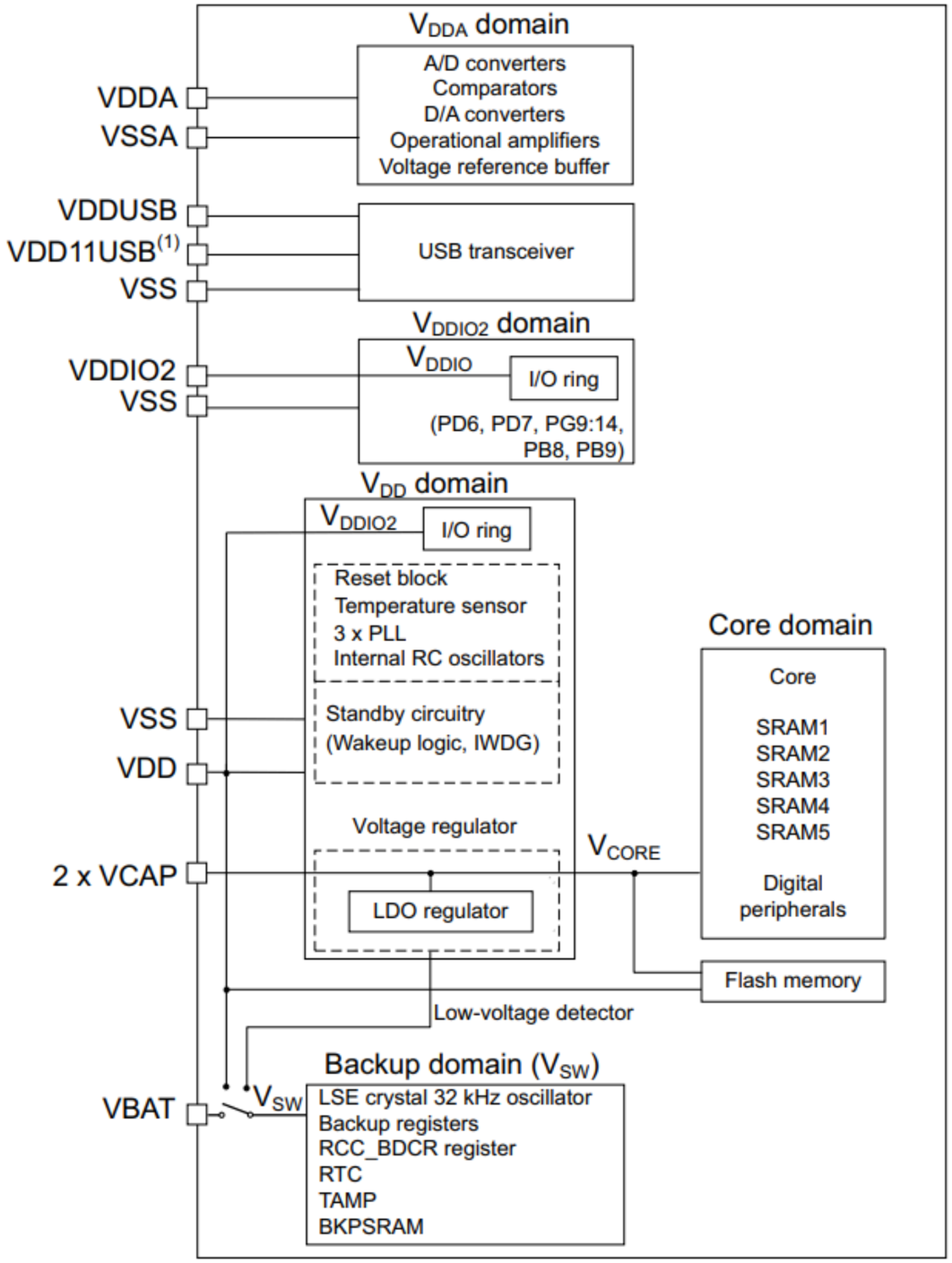
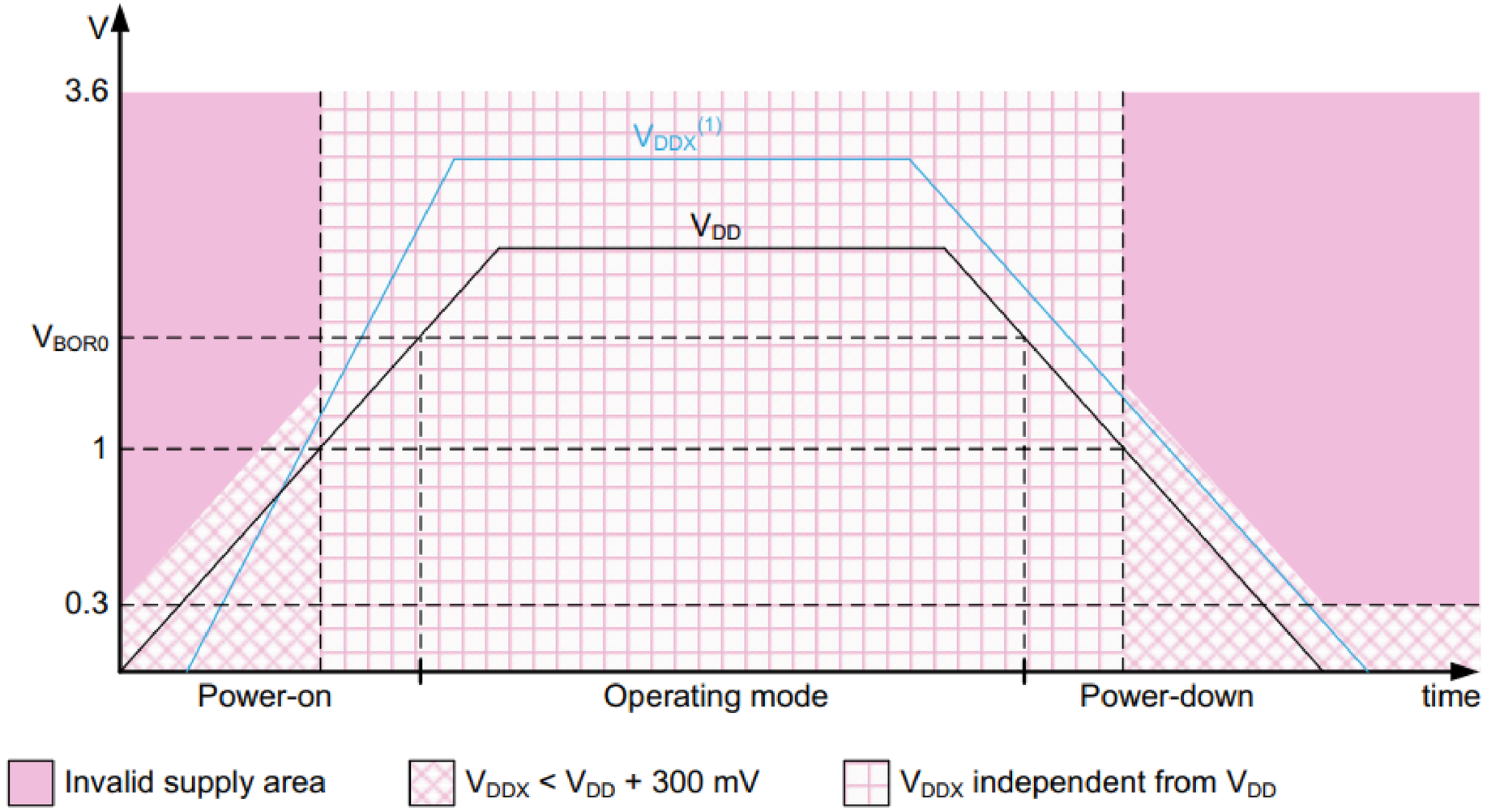

2 channels, 1 compl. channel, BKIN as AF
TIM15 16-bit
1 channel, 1 compl. channel, BKIN as AF
TIM16 16-bit
1 channel, 1 compl. channel, BKIN as AF
TIM17 16-bit
RX, TX, CK,CTS, RTS_DE as AF
smcard irDA USART1
MOSI, MISO, SCK, NSS as AF
SPI1/I2S1
MOSI, MISO, SCK, NSS as AF
SPI4
MOSI, MISO, SCK, NSS as AF
SPI6
MCLK_A, SD_A, FS_A, SCK_A, MCLK_B, SD_B, FS_B, SCK_B as AF
SAI1
AUDIOCLK as AF
MCLK_A, SD_A, FS_A, SCK_A, MCLK_B, SD_B, FS_B, SCK_B as AF
SAI2
APB2
Temperature monitoring
OSC32_IN, OSC32_OUT
VBAT
XTAL 32k
RTC_OUT1, RTC_OUT2, RTC_REFIN, RTC_TS
RTC
RTC_OUT[8:1], RTC_IN[8:1]
TAMP
VDDA
VREF+
VREF buffer
IN1, IN2, CH1, CH2, ETR as AF
LPTIM1
SCL, SDA, SMBA as AF
I2C3/SMBUS
SCL, SDA, SMBA as AF
I2C4/SMBUS
RX, TX, CTS, RTS_DE as AF
LPUART1
MOSI, MISO, SCK, NSS as AF
SPI5
SBS
IN1, IN2, CH1, CH2, ETR as AF
LPTIM3
IN1, ETR as AF
LPTIM4
IN1, IN2, CH1, CH2, ETR as AF
LPTIM5
IN1, IN2, CH1, CH2, ETR as AF
LPTIM6
SCL, SDA
I3C2
IN, OUT
PLAY1
APB3 250 MHz
AHB/APB3
DTS
WWDG
IWDG
TIM6 16-bit
TIM7 16-bit
SPI3/I2S3
MOSI, MISO, SCK, NSS as AF
I2C1/SMBUS
SCL, SDA, SMBA as AF
I2C2/SMBUS
SCL, SDA, SMBA as AF
FDCAN1
FDCAN2
FDCAN3
FIFO
TX, RX as AF
TX, RX as AF
TX, RX as AF
UCPD1
PHY
CC1, DBCC1, CC2, DBCC2, FRSCC1, FRSCC2 as AF
LPTIM2
IN1, IN2, CH1, CH2, ETR as AF
TIM12 16-bit
2 channels, ETR as AF
TIM13 16-bit
1 channel, ETR as AF
TIM14 16-bit
1 channel, ETR as AF
UART7
RX, TX, CTS, RTS_DE as AF
UART8
RX, TX, CTS, RTS_DE as AF
UART9
RX, TX, CTS, RTS_DE as AF
USART10
smcard irDA
RX, TX, CK, CTS, RTS_DE as AF
USART11
smcard irDA
RX, TX, CK, CTS, RTS_DE as AF
UART12
RX, TX, CTS, RTS_DE as AF
HDMI-CEC
CEC
I3C1
SCL, SDA
ITF
VDDA
COMP1
COMP2
INN, INP, OUT
INN, INP, OUT
ITF
VDDA
OPAMP1
VINN, VINP, VOUT
VDD power domain
VDDA power domain
VDDUSB power domain
VBAT power domain

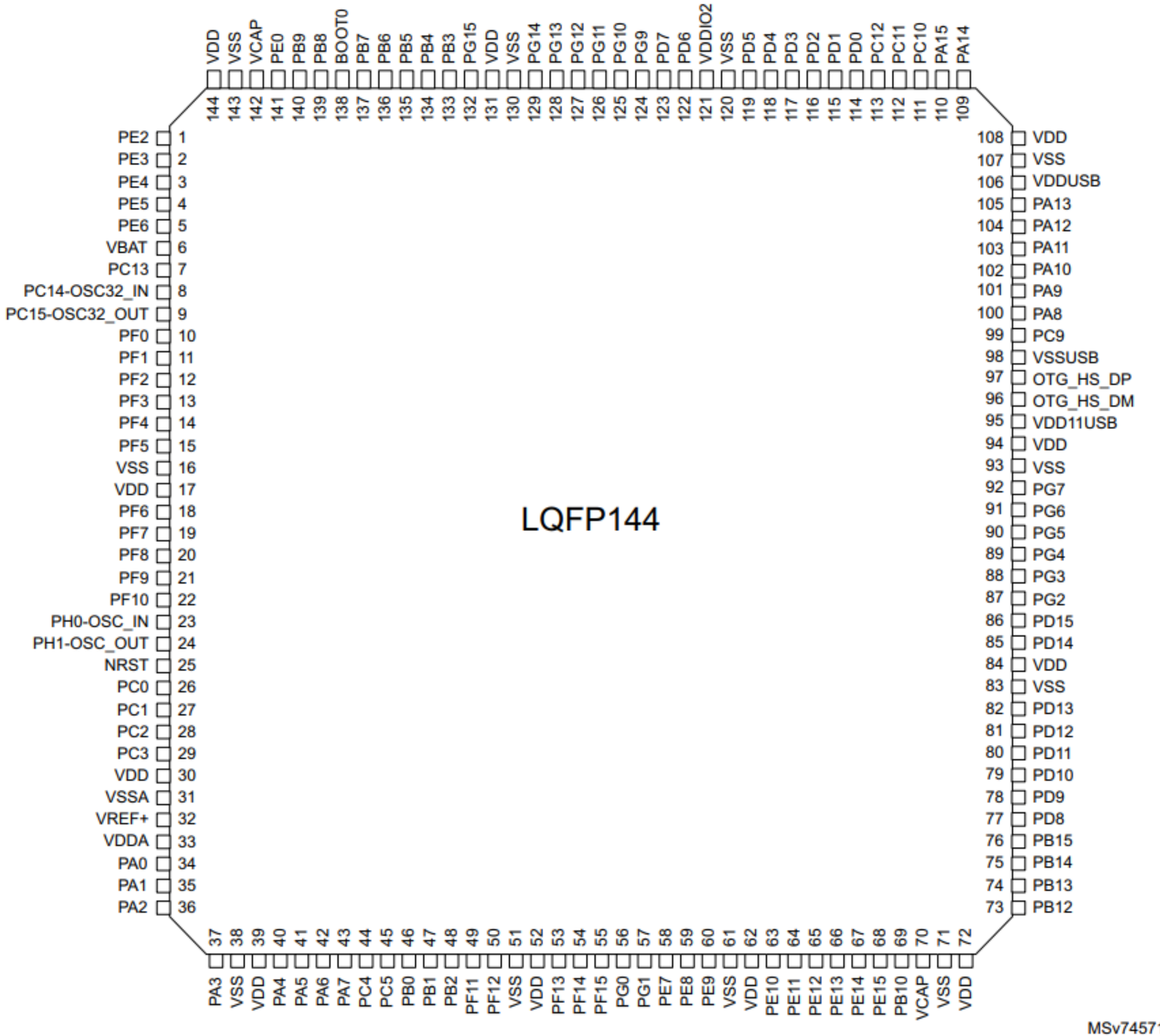
DIAGRAMA DA ALIMENTAÇÃO DE ENERGIA DO MCU



SEQUÊNCIA DE ALIMENTAÇÃO ELÉTRICA (POWER-UP/DOWN)



MAPEAMENTO DE PINOS



MAPEAMENTO DE PINOS

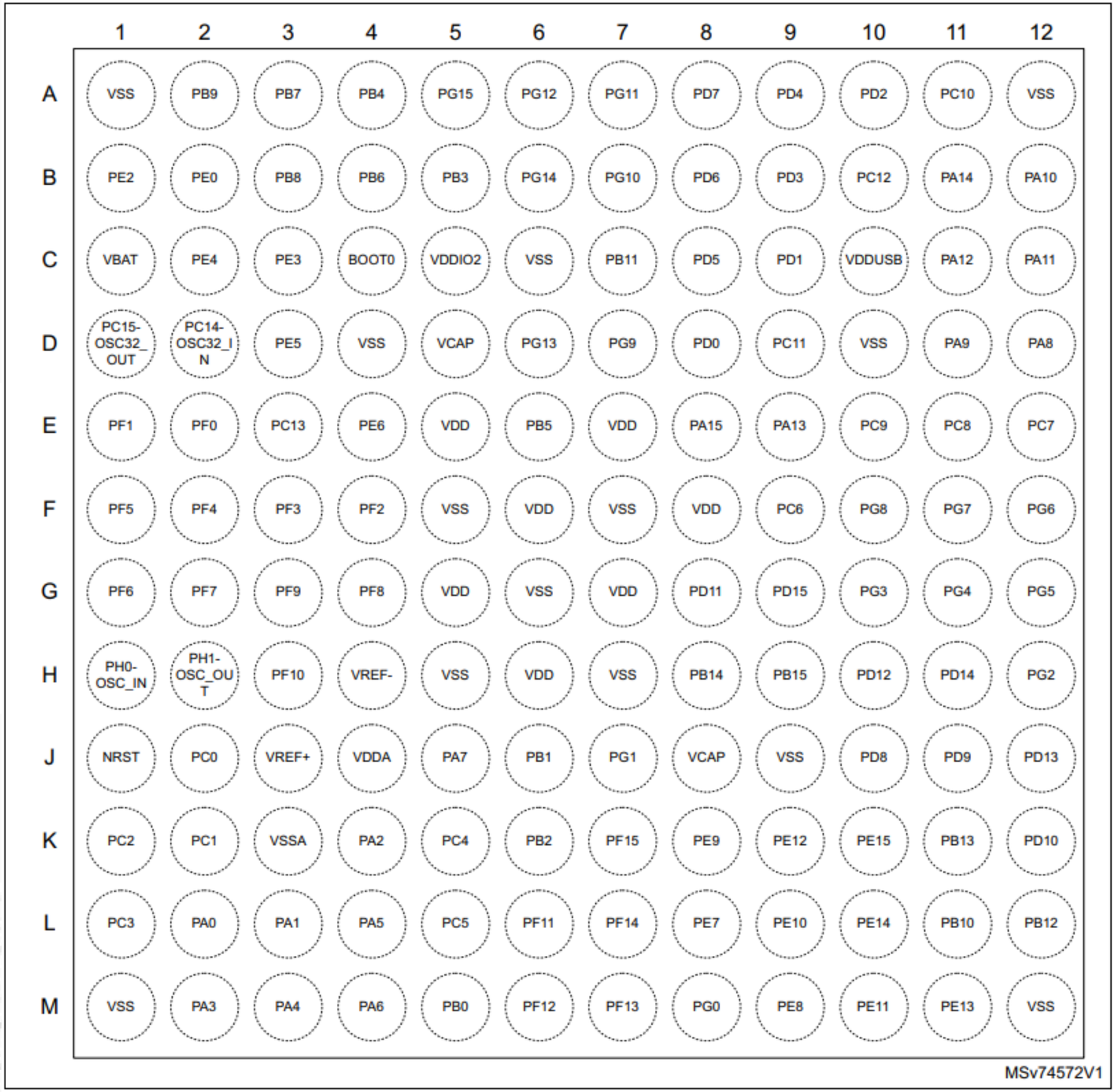


DIAGRAMA DE TEMPORIZAÇÃO CA DE FONTE DE CLOCK EXTERNO DE ALTA/BAIXA VELOCIDADE

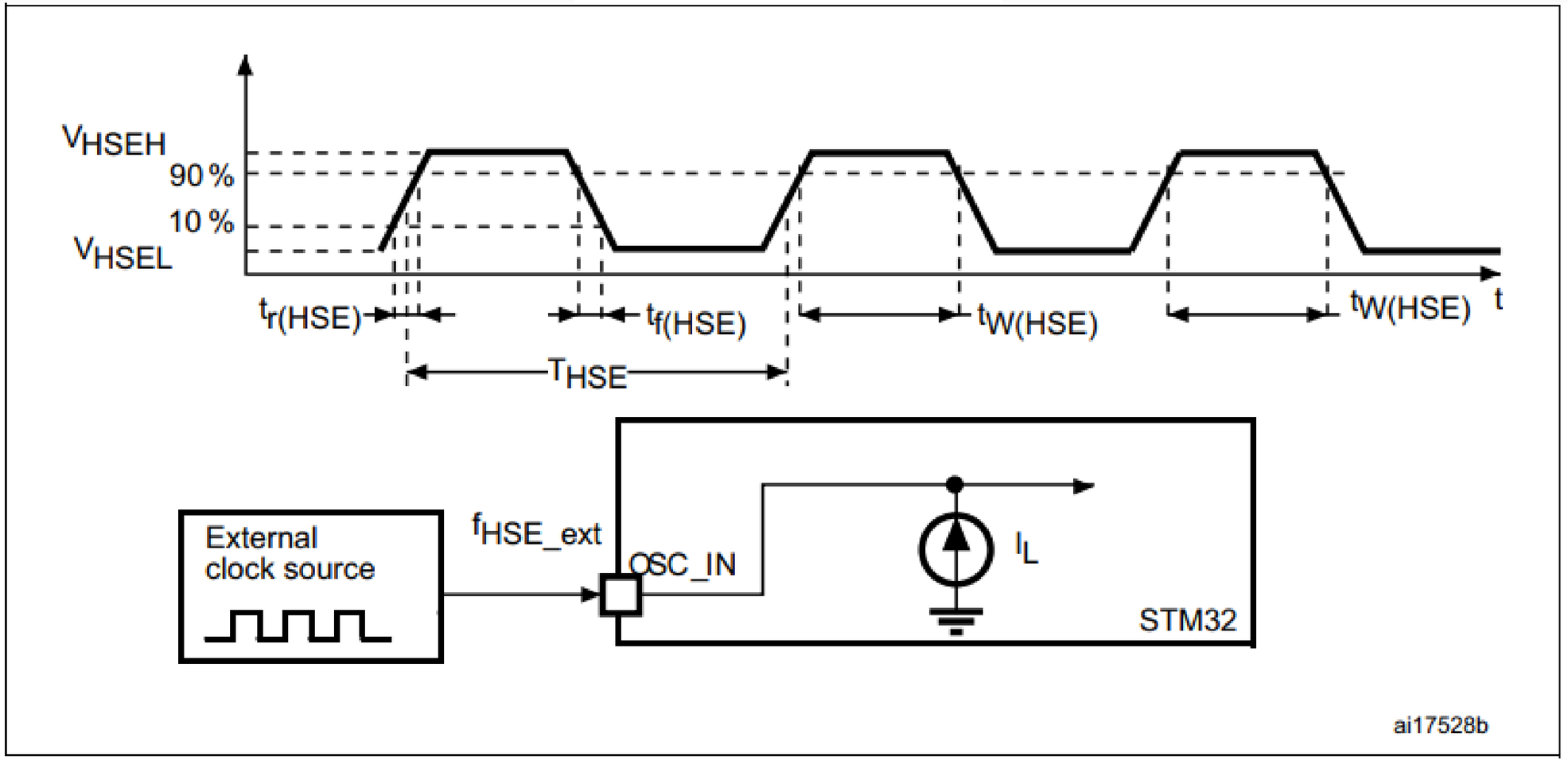


DIAGRAMA DE TEMPO SPI - MODO ESCRAVO E CPHA = 0

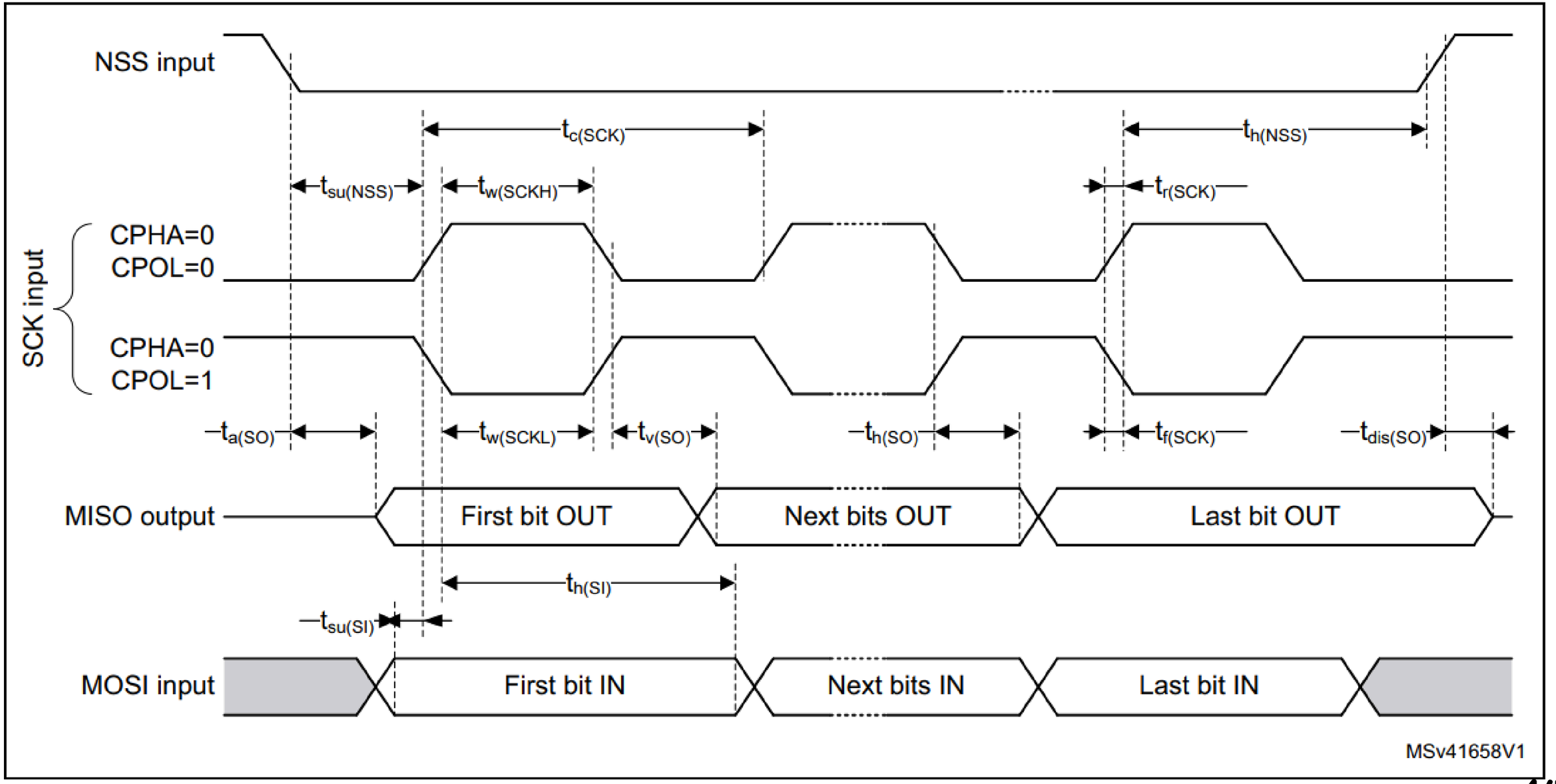


DIAGRAMA DE TEMPO SPI - MODO ESCRAVO E CPHA = 1

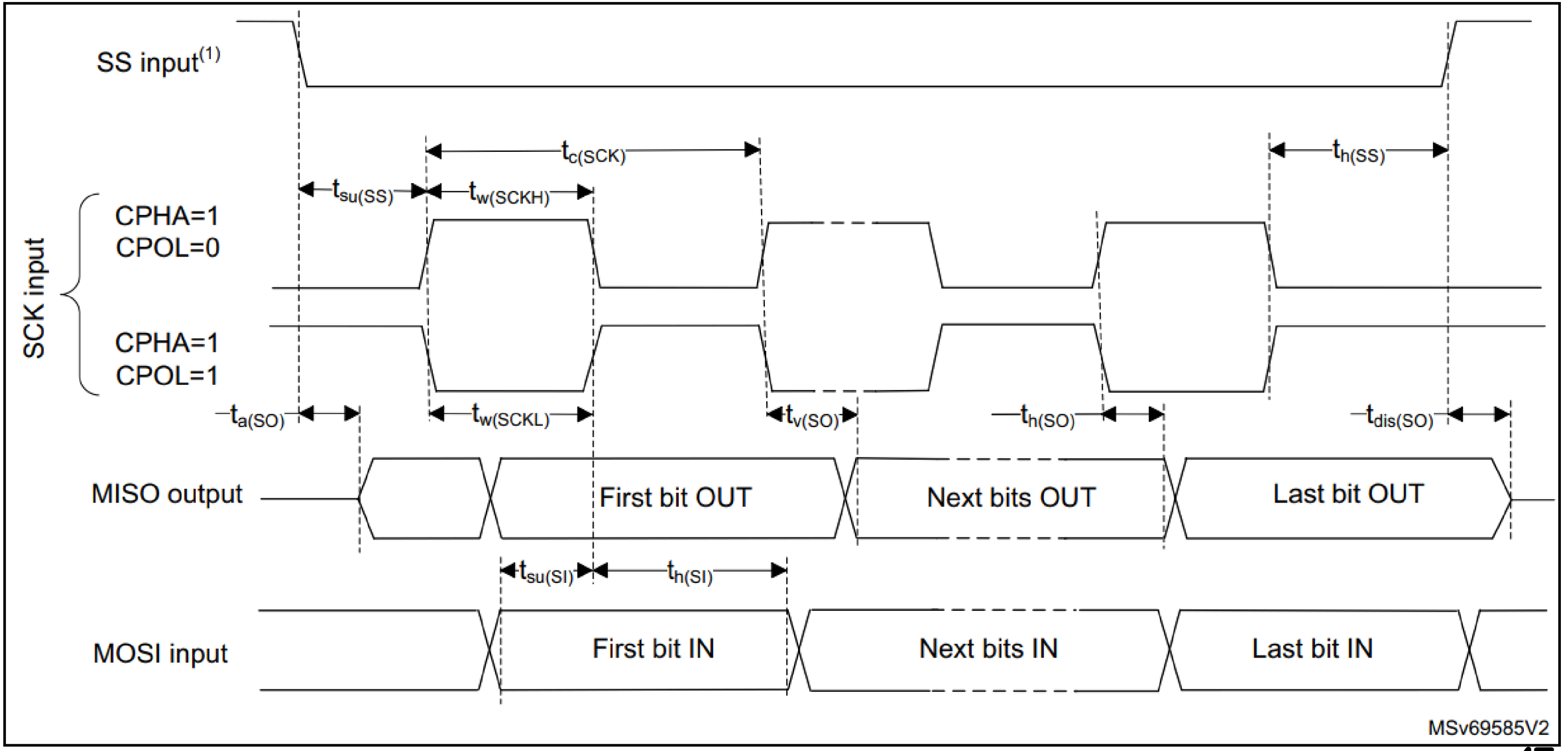


DIAGRAMA DE TEMPO USART - MODO MESTRE

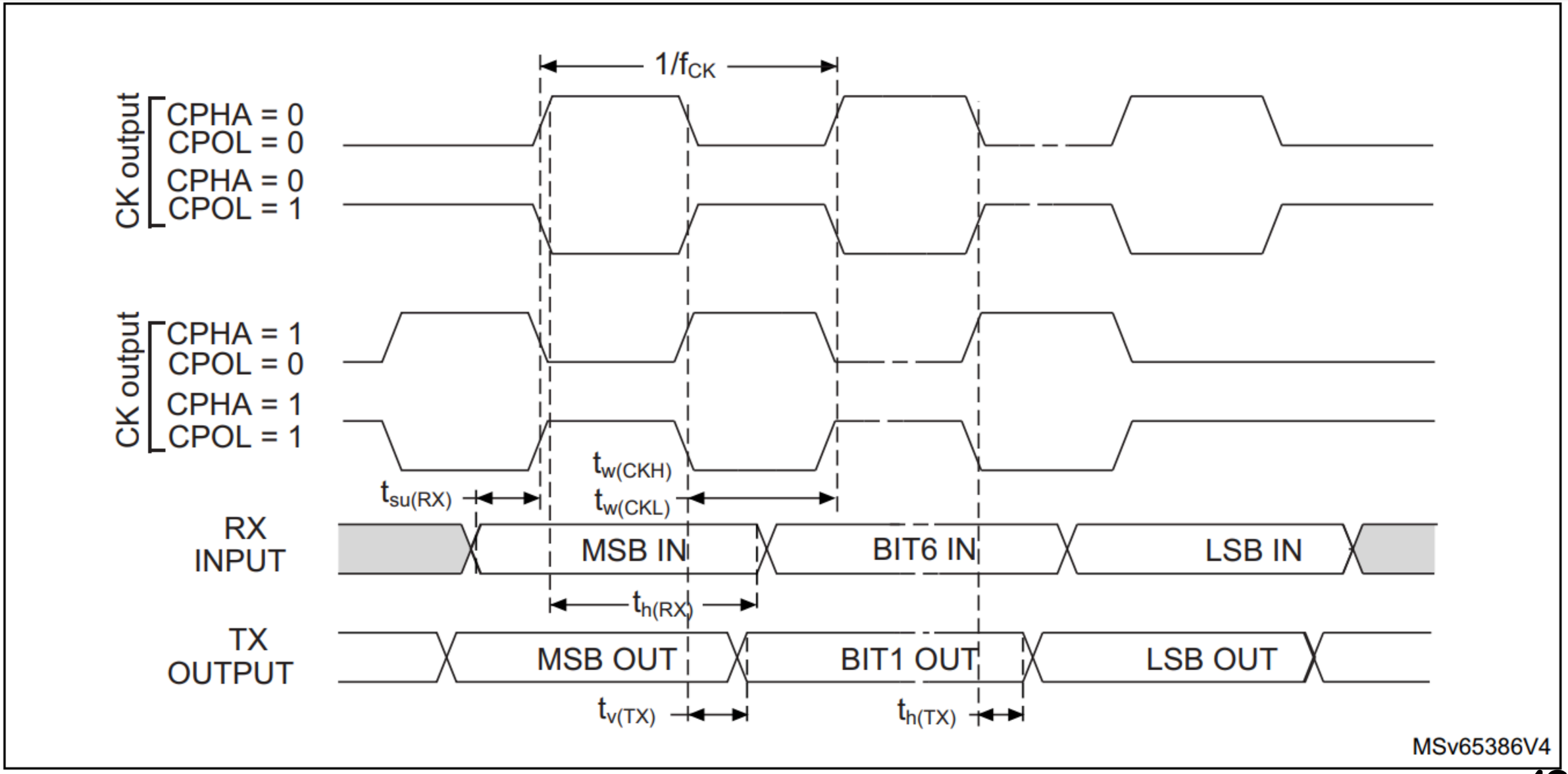


DIAGRAMA DE TEMPO USART - MODO ESCRAVO

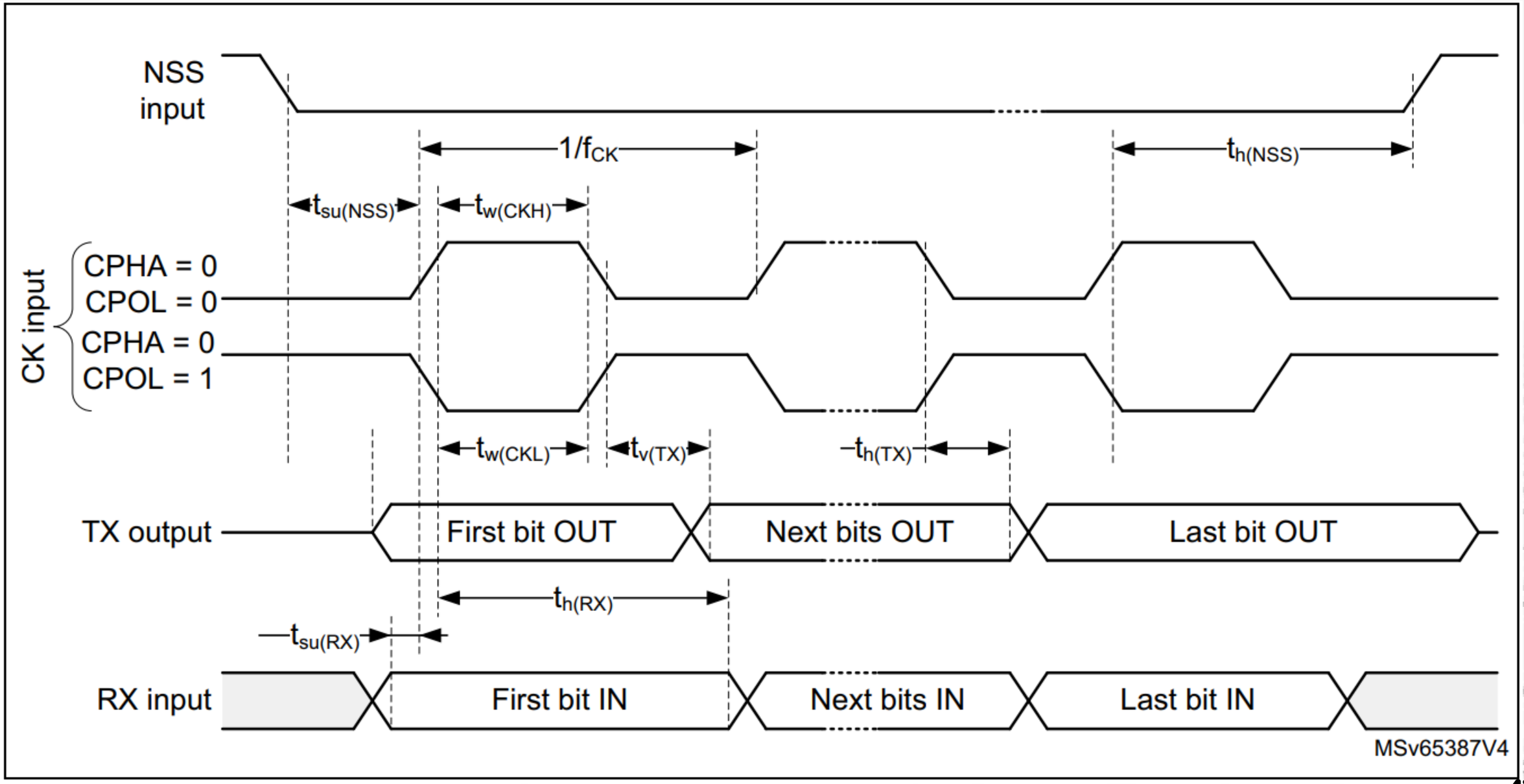
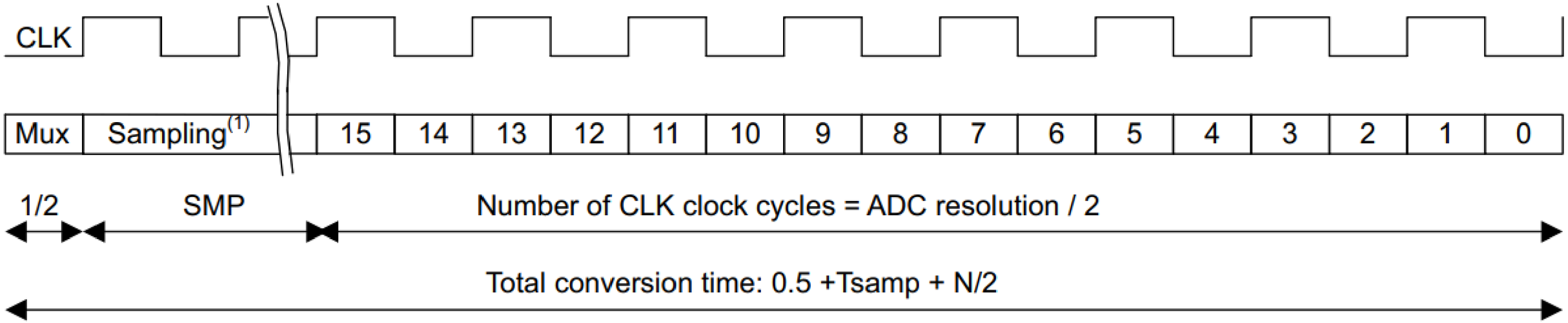


DIAGRAMA DE TEMPO DO ADC



1. The sampling time defines the minimum sampling clock cycles (SMP) to be programmed in the ADC (refer to the product reference manual for details).

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MICROPROCESSADOR

- **QCS8275 TAMBÉM CONHECIDO COMO IQ-8275**

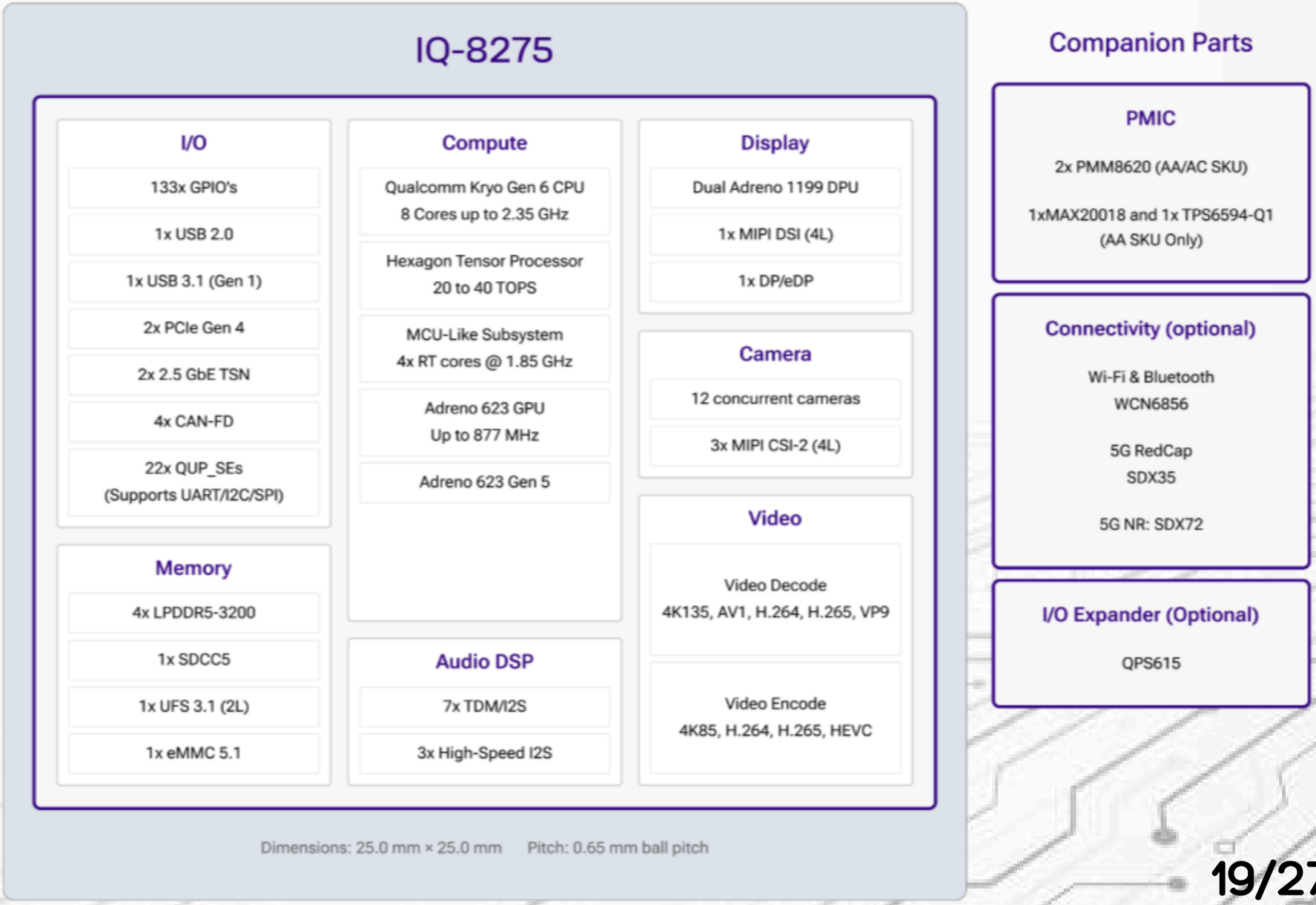
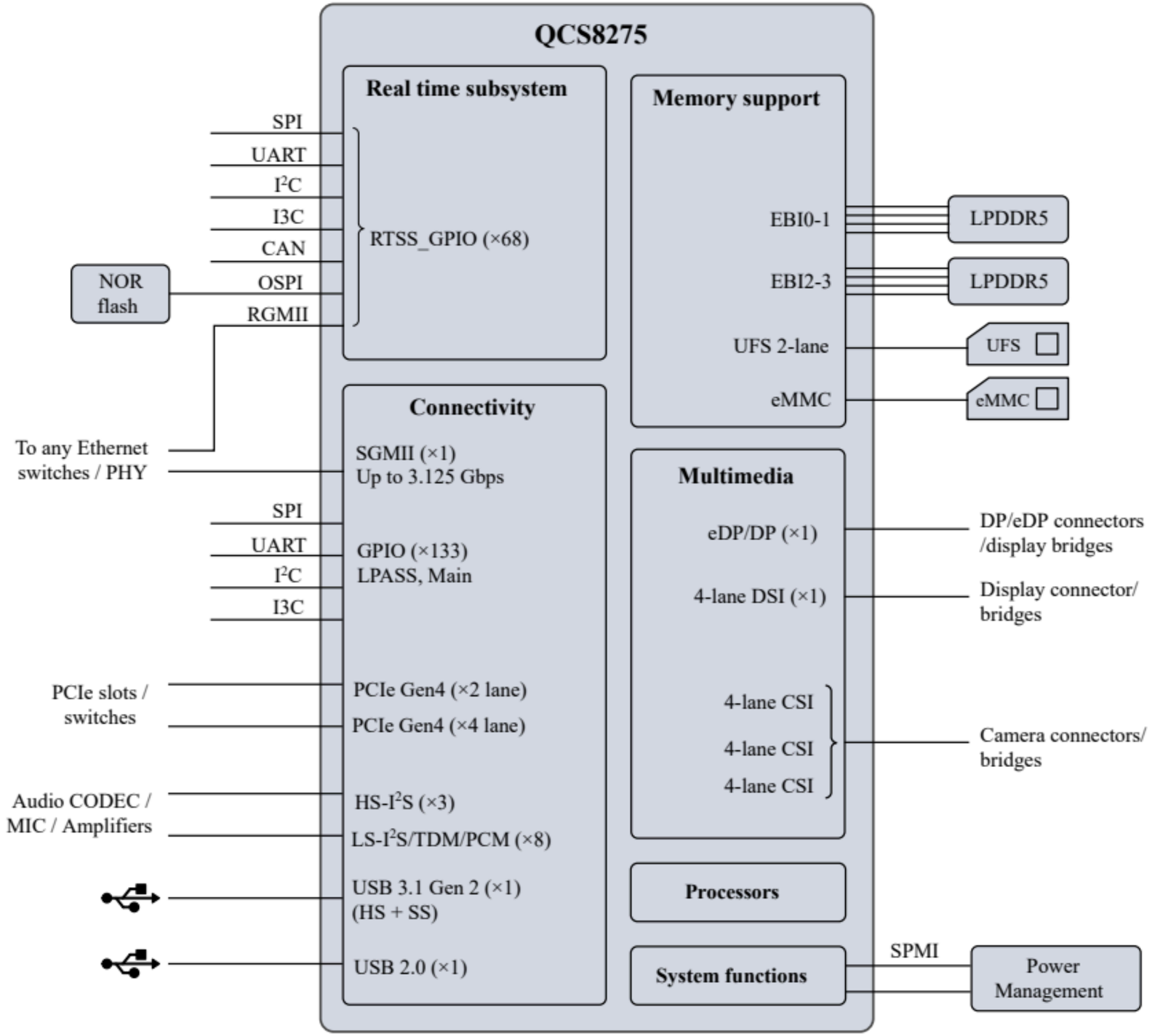
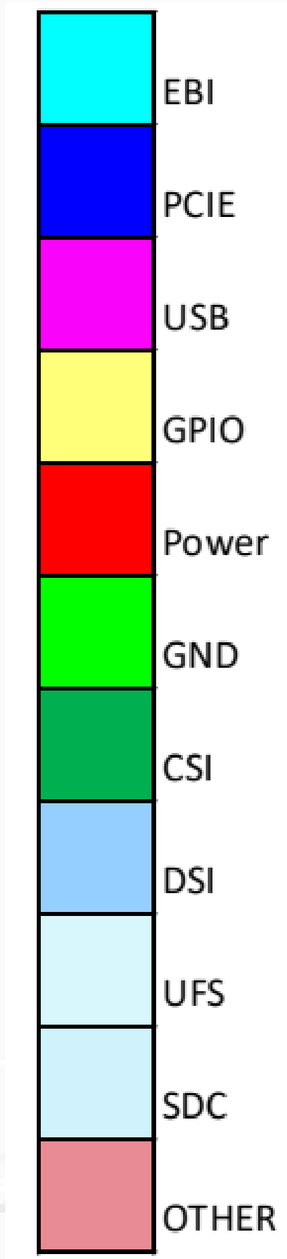


DIAGRAMA DE BLOCO DE ALTO NÍVEL



MAPEAMENTO DE PINOS



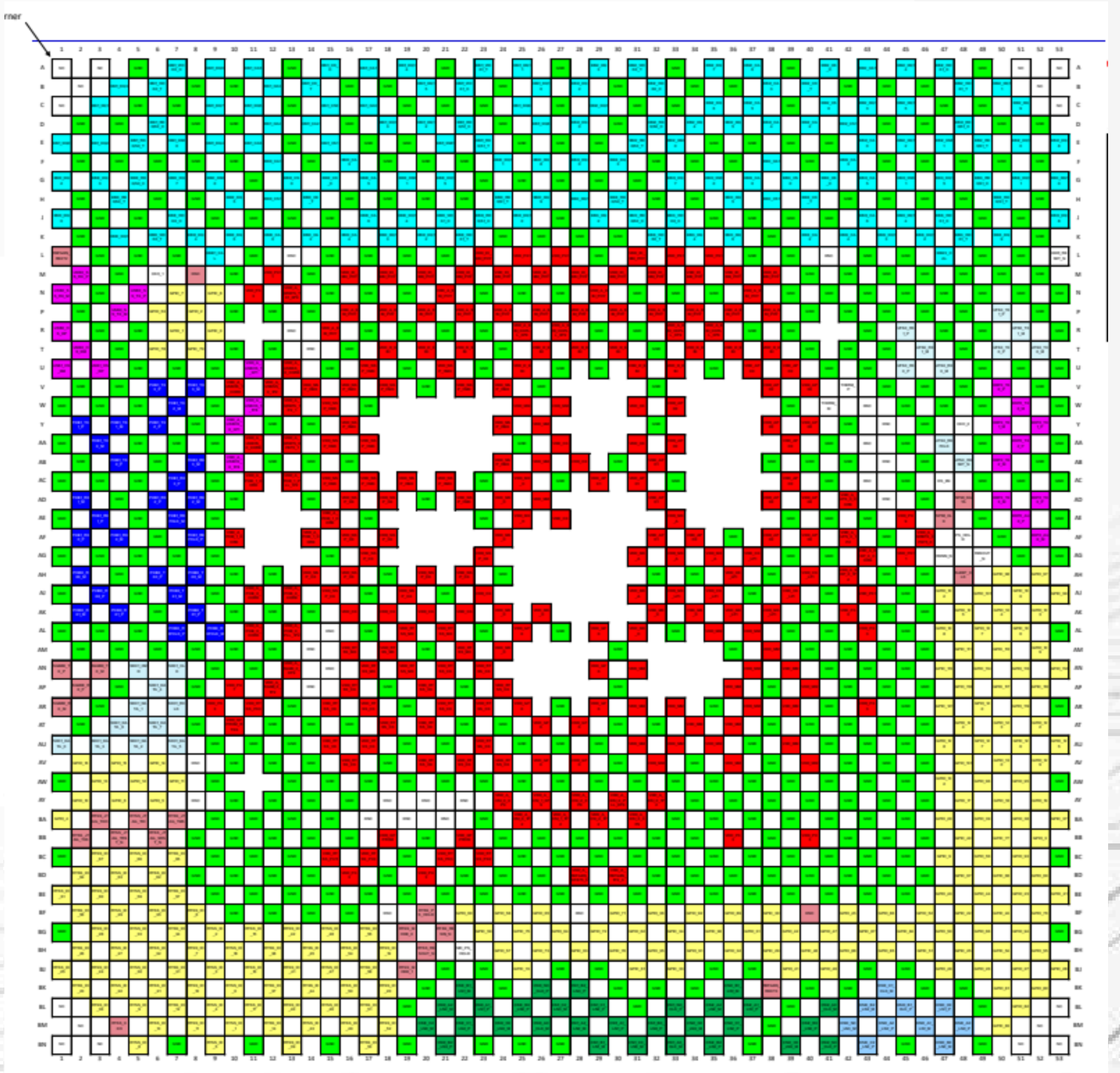


DIAGRAMA DE TEMPO OCTA-SPI SDR

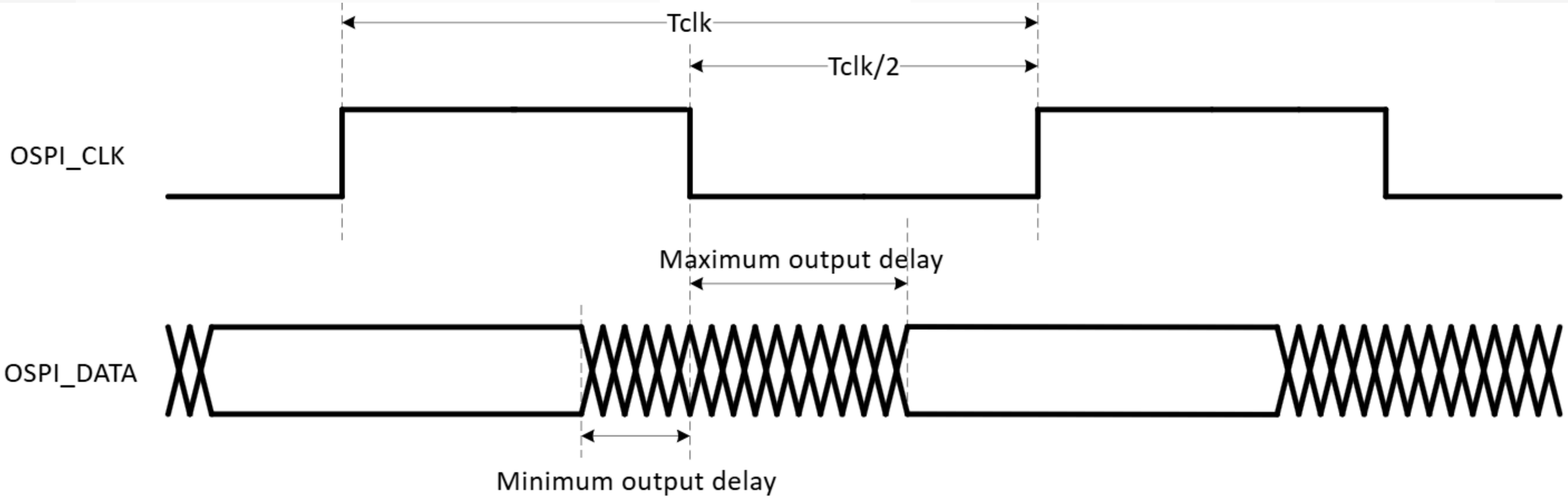


DIAGRAMA DE TEMPO OCTA-SPI DDR

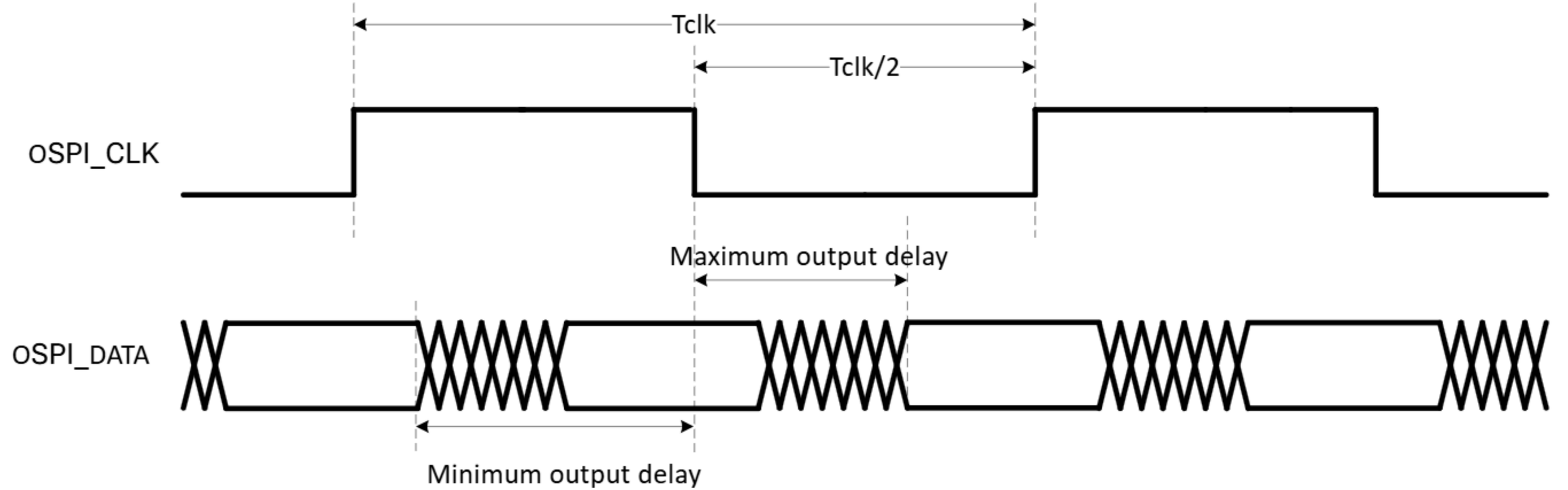


DIAGRAMA DE TEMPO RGMII TX E RX

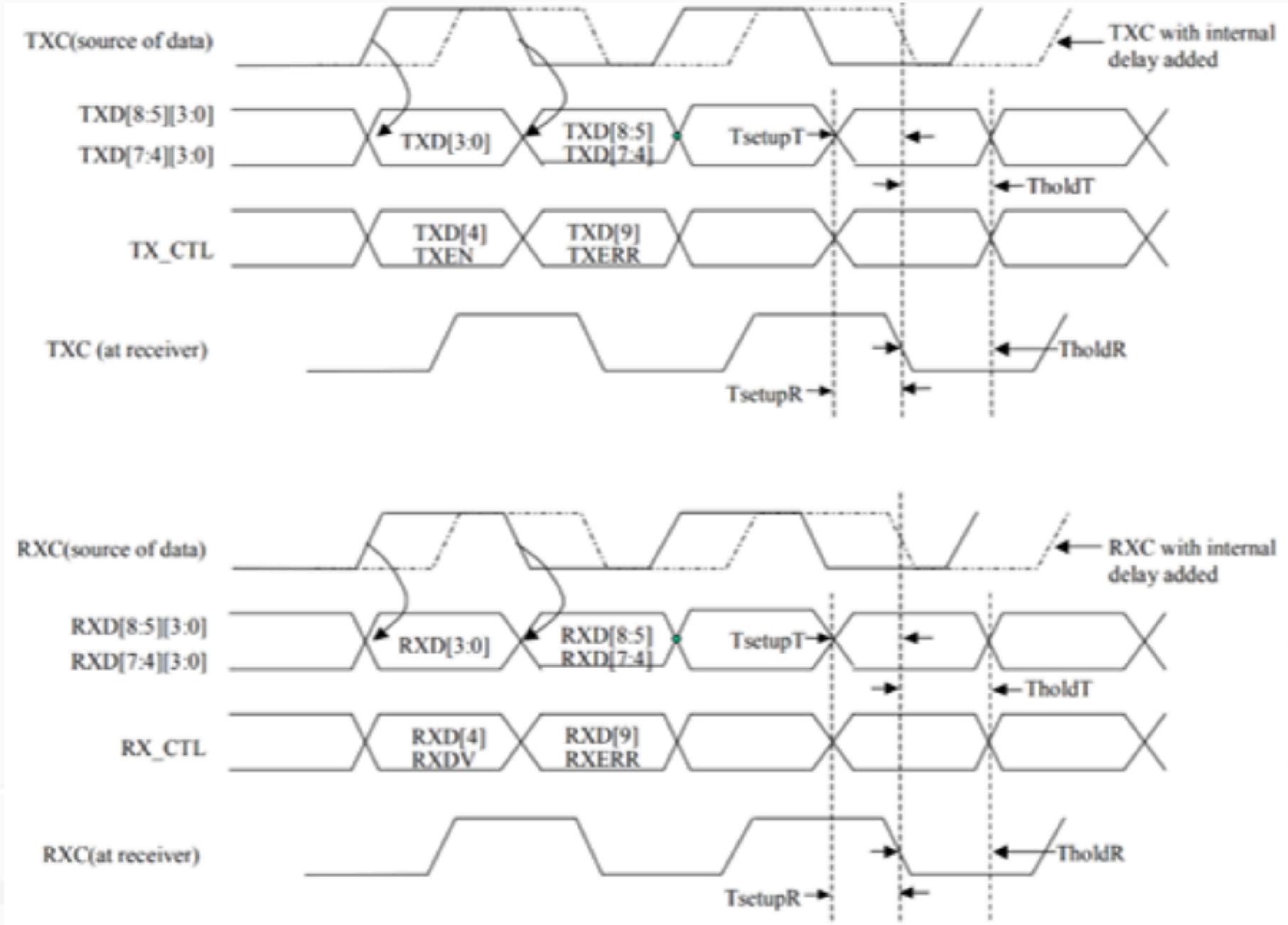
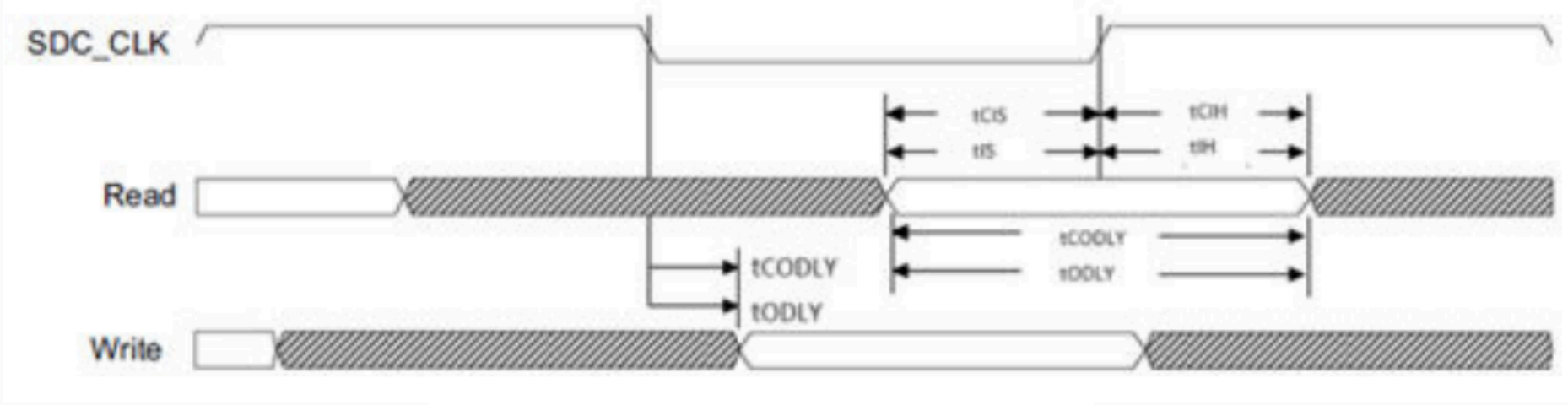
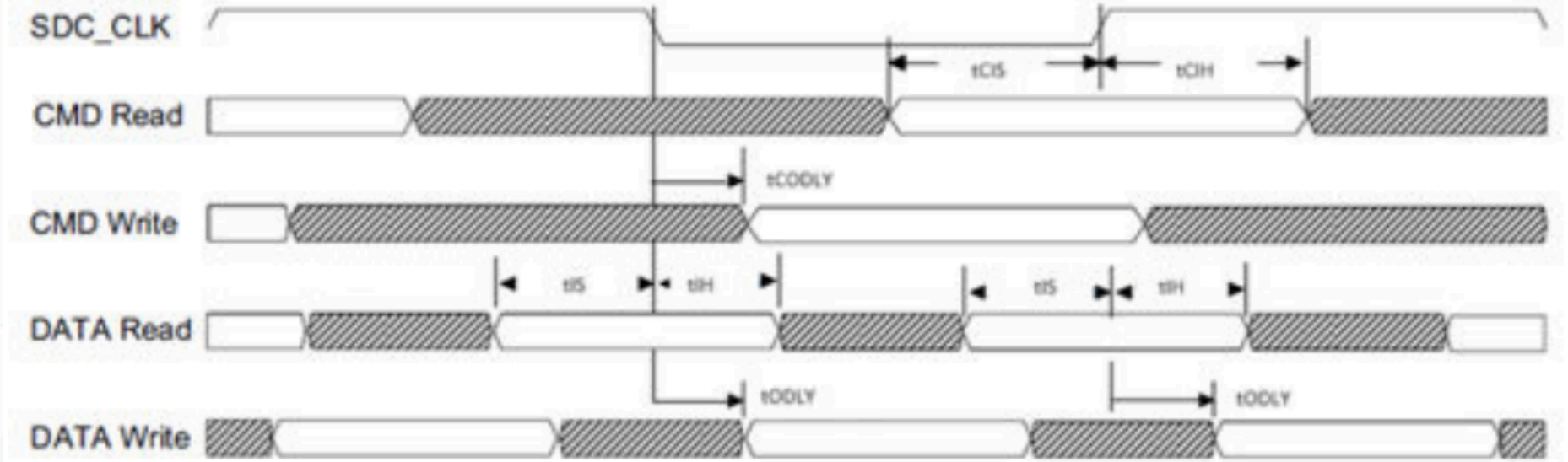


DIAGRAMA DE TEMPO SDC

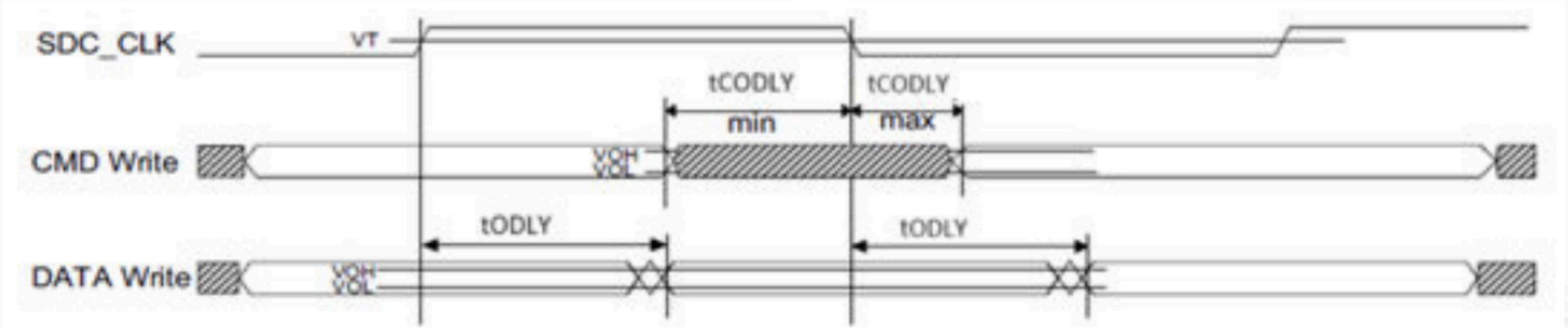


SDR

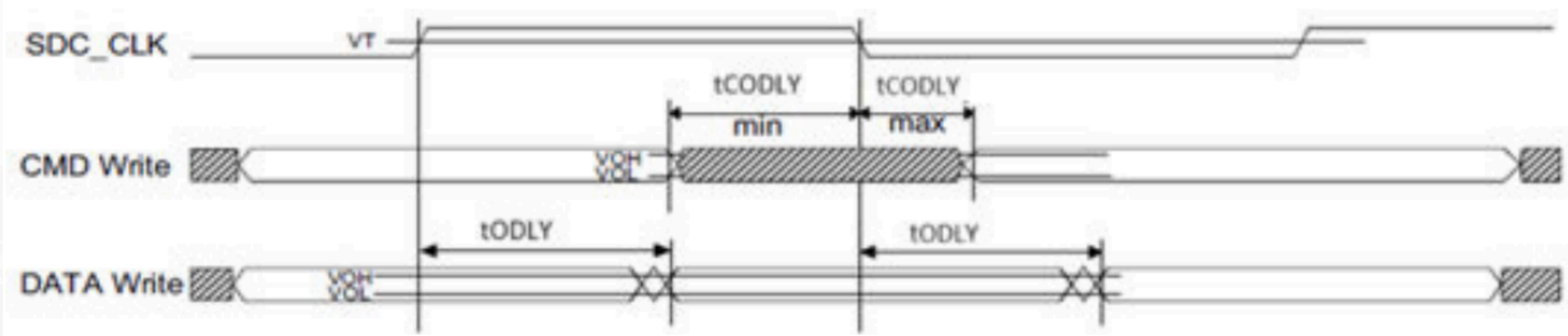


DDR

DIAGRAMA DE TEMPO DO MODO HS400 I/O



IN



OUT

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CONCLUSÃO

- A CPU e MCU em conjunto permite distribuir tarefas conforme as necessidades de processamento e controle.

- A arquitetura computacional mantém princípios tradicionais de processamento, mas incorpora melhorias em desempenho.